1. States, matrices and entangling gates

2023 Q3; all pulse and protocol questions

Universal exam procedure: (1) State the basis and qubit ordering; (2) identify the physical subspace or operator; (3) apply it to each relevant basis input, preserving relative phases; (4) normalise or assemble the matrix; (5) check unitarity, probabilities and whether any ancilla returns to its initial state.

A. Translate a Hamiltonian into a gate

For a time-independent closed-system Hamiltonian,

$$U(t) = \mathrm{e}^{-iHt/\hbar}.$$

If $A^2 = \mathbb{I}$ (a Pauli matrix or Pauli product),

$$\boxed{\mathrm{e}^{-i\theta A} = \cos\theta\,\mathbb{I} - i\sin\theta\,A.}$$

Method: expand H; move any term proportional to $\mathbb{I}$ into a global phase; identify $A^2 = \mathbb{I}$; substitute into the identity; *apply to input states rather than guessing the gate*. Keep the global phase if asked for the exact unitary, omit it only when identifying physical gate action.

B. Worked pattern: the 2023 YY Hamiltonian

2023 Q3A1–2: $H = \frac{\hbar g}{2}(Y_1 + Y_2)^2$. Since $Y_i^2 = \mathbb{I}$ and $[Y_1, Y_2] = 0$,

$$H = \hbar g(\mathbb{I} + Y \otimes Y),$$

$$U = \mathrm{e}^{-igt}\{\cos(gt)\mathbb{I} - i\sin(gt)Y \otimes Y\}.$$

The overall e^{-igt} is a global phase. Using $Y|0\rangle = i|1\rangle$,

$$U|00\rangle \sim \cos(gt)|00\rangle + i\sin(gt)|11\rangle.$$

Bell-state condition: equal amplitudes $\Rightarrow \cos(gt) = \sin(gt)$, so

$$\boxed{t_{\mathrm{ent}} = \frac{\pi}{4g}, \quad |\psi\rangle \sim \frac{|00\rangle + i|11\rangle}{\sqrt{2}}.}$$

Test: the reduced state of either qubit is $\mathbb{I}/2$, confirming maximal entanglement.

C. Convert an entangler to CNOT

2023 Q3B; 2025 Q2A5. Use local rotations to change the Pauli axis, then transform CZ into CNOT:

$$R_x(\pi/2)\,Y\,R_x(-\pi/2) = Z,$$

$$\mathrm{e}^{-i\pi YY/4} \xrightarrow{R_x \otimes R_x} \mathrm{e}^{-i\pi ZZ/4}.$$

For $S = \mathrm{diag}(1, i)$,

$$(S^\dagger \otimes S^\dagger)\mathrm{e}^{-i\pi ZZ/4} = \mathrm{e}^{-i\pi/4}\mathrm{CZ},$$

$$\boxed{\mathrm{CNOT}_{A\to B} = (\mathbb{I} \otimes H)\mathrm{CZ}(\mathbb{I} \otimes H).}$$

Procedure: show axes $YY \to ZZ$; remove local phase gates; put Hadamards around *target* to turn controlled phase into controlled flip.

D. Qubit rotations and pulse areas

For a resonantly driven qubit,

$$H = \frac{\hbar\Omega}{2}\,\hat{\boldsymbol{n}}\cdot\boldsymbol{\sigma}, \qquad \theta = \Omega t,$$

$$R_{\hat{\boldsymbol{n}}}(\theta) = \cos\tfrac{\theta}{2}\mathbb{I} - i\sin\tfrac{\theta}{2}\hat{\boldsymbol{n}}\cdot\boldsymbol{\sigma}.$$

A π pulse transfers population in the addressed two-state subspace (phase $-i$ for a standard x pulse); a 2π pulse returns the state with phase -1. For a general drive

$$H_{\mathrm{rot}} = \tfrac{\hbar}{2}(\Omega X + \delta Z), \quad \Omega_R = \sqrt{\Omega^2 + \delta^2}.$$

Watch the factor of two: if $H = \hbar t_c X$ (rather than $\hbar\Omega X/2$), then $\theta = 2t_c t$.

E. Work with qubit states and density matrices

$$|\psi\rangle = \cos\tfrac{\theta}{2}|0\rangle + \mathrm{e}^{i\phi}\sin\tfrac{\theta}{2}|1\rangle,$$

$$\rho = \tfrac{1}{2}(\mathbb{I} + r_x X + r_y Y + r_z Z)$$

$$= \tfrac{1}{2}\begin{pmatrix} 1 + r_z & r_x - ir_y \\ r_x + ir_y & 1 - r_z \end{pmatrix}.$$

Reading a matrix: diagonal → populations; off-diagonals → coherence. Bloch components $r_j = \mathrm{Tr}(\rho\sigma_j)$; pure states $|\boldsymbol{r}| = 1$, mixed states $|\boldsymbol{r}| < 1$.

$$\mathrm{Tr}(\rho^2) = \tfrac{1}{2}(1 + |\boldsymbol{r}|^2), \quad \langle A\rangle = \mathrm{Tr}(\rho A).$$

For a bipartite state, discard an inaccessible system by $\rho_B = \mathrm{Tr}_A\,\rho_{AB}$. For a Bell state, ρ_{AB} is pure but $\rho_A = \rho_B = \mathbb{I}/2$.

F. Recognise the physical answer being sought

Question wording	What to show
"Determine unitary"	$U = \exp(-iHt/\hbar)$, basis, phases
"Maximally entangled"	two equal Schmidt coefficients
"Gate matrix"	transform *all* basis vectors; outputs are columns
"CNOT from CZ"	Hadamards on the target
"Fastest gate"	maximise available angular frequency; $t = \theta/\Omega_R$

Common mistake: $Y \otimes Y$ is not $Y \otimes \mathbb{I} + \mathbb{I} \otimes Y$. Distinguish the Hamiltonian's stated Pauli product from its square and include all global factors in an exact exponential.

2. Stabiliser codes: solve any unfamiliar code

2023 Q2; 2024 Q1; 2025 Q1

Six-step workflow for every past-paper QEC problem: Check generators commute → calculate k → compute syndromes → test correctability (including degeneracy) → find X_L, Z_L → prove the minimum distance d. This is a method, not a code to memorise.

A. Commutation without large matrices

At a qubit position, distinct nonidentity Paulis (X/Y, $X/Z, Y/Z$) *anticommute*; equal Paulis or any I commute. Two tensor-product Paulis obey

$$PQ = (-1)^m QP,$$

where m is the *number of anticommuting positions*. Even $m \to$ commute; odd $m \to$ anticommute.

$$\begin{aligned} ZIZ \quad &\text{vs} \quad XIY: \ m = 2 \ \checkmark \\ ZIZ \quad &\text{vs} \quad XII: \ m = 1 \ \times \end{aligned}$$

For n qubits and r independent, commuting generators (with $-I$ excluded):

$$\boxed{k = n - r.}$$

Why? Each independent $+1$ stabiliser halves the 2^n-dimensional Hilbert space; the code has dimension 2^{n-r}.

B. Syndrome table, one physical error at a time

For an error E, define $b_i = 0$ if $[E, S_i] = 0$ and $b_i = 1$ if $\{E, S_i\} = 0$. The measured eigenvalue is $(-1)^{b_i}$. *Calculate each b_i by counting overlaps.*

Example: $S_1 = Z_1Z_2$, $S_2 = Z_2Z_3$ for the three-qubit bit-flip code:

E	S_1	S_2	syndrome b_1b_2
I	$+$	$+$	00
X_1	$-$	$+$	10
X_2	$-$	$-$	11
X_3	$+$	$-$	01
Z_1	$+$	$+$	00

The two checks identify any *single bit flip* but cannot detect Z_1: this is not a general one-error quantum code.

C. Detection vs correction; resolve collisions

Detect an error if its syndrome is nonzero (or it is a stabiliser acting trivially). To **correct an entire set** $\{E_a\}$, comparing two possible errors matters:

1. Different syndromes → decoder can distinguish them.
2. Same syndrome → calculate $E_a^\dagger E_b$.
3. If $E_a^\dagger E_b \in \mathcal{S}$, the collision is *harmless degeneracy*. If it lies in $\mathcal{N}(\mathcal{S}) \setminus \mathcal{S}$, it is an uncorrectable logical ambiguity.

General condition: $PE_a^\dagger E_b P = c_{ab} P$, with P the codespace projector.

D. Two actual collisions to recognise

2023 Q2: $\mathrm{syn}(Z_1) = \mathrm{syn}(Z_3)$, but $Z_1Z_3 = S_1$ (a stabiliser). Either correction has the same logical action: *correctable.*

2024 Q1: $\mathrm{syn}(X_1) = \mathrm{syn}(X_2)$, but $X_1X_2 = X_L$ is a nontrivial logical operation. Wrong recovery changes the encoded state: *not fully correctable.*

E. Find logical Paulis and code distance

Candidate logical L must commute with *every* stabiliser but not belong to $\mathcal{S}$. Select X_L, Z_L with

$$[L, S_i] = 0, \quad X_L Z_L = -Z_L X_L.$$

Multiplication by a stabiliser gives an equivalent logical representative. Weight $\mathrm{wt}(L)$ counts non-I factors. The distance is

$$\boxed{d = \min_{L \in \mathcal{N}(\mathcal{S}) \setminus \mathcal{S}} \mathrm{wt}(L), \qquad t = \left\lfloor \frac{d-1}{2} \right\rfloor.}$$

To prove d**:** (1) show a particular logical operator of weight $w \Rightarrow d \le w$; (2) rule out all lower-weight nontrivial logicals (often by showing correction of all one-qubit Pauli errors) $\Rightarrow d \ge w$.

F. Historical code results (reusable checks)

Paper	$[[n, k, d]]$	X_L example	Z_L example
2023	$[[6, 1, 3]]$	$XZXZII$	$XYYXII$
2024	$[[5, 1, 2]]$	XXIII	$ZIIZI$
2025	$[[9, 1, 3]]$	$Z_1Z_4Z_7$	$X_1X_2X_3$

2025 Shor-type code: six Z-pair checks identify a bit flip within a 3-qubit block. Two six-body X checks locate a phase-flip *block*. Different Z errors within the same block differ by stabilisers, so they are equivalent and correctable. Y triggers both patterns.

G. CSS and transversal-gate cue

In CSS codes, *Z checks detect X errors*, while *X checks detect Z errors*. The Steane code is $[[7, 1, 3]]$; it corrects one arbitrary physical Pauli error. A transversal CNOT is formed by applying physical CNOTs between *corresponding* qubits of two blocks; show its action on X_L, Z_L to establish the logical gate.

Exam trap: all 15 weight-one Paulis can have nonzero syndromes and the code can still fail to *correct* one error. The 2024 code is precisely this example: $d = 2$, so $t = 0$ for arbitrary noise.

3. Physical gates: trapped ions and Rydberg atoms

2023 Q1; 2024 Q3; 2025 Q2

Physical-gate recipe: State which electronic/motional transition is resonant; identify dark (uncoupled) branches; track each basis state through *each* pulse; retain factors $-i$ and -1; check the bus/auxiliary system is restored; read off the computational unitary.

A. Ions: sidebands and phonon factors

Write the state as $|\text{ion}, n\rangle$, including phonon number n.

Laser	Transition	Rate
Carrier	$\|g,n\rangle \leftrightarrow \|e,n\rangle$	Ω
Red	$\|g,n\rangle \leftrightarrow \|e,n-1\rangle$	$\eta\Omega\sqrt{n}$
Blue	$\|g,n\rangle \leftrightarrow \|e,n+1\rangle$	$\eta\Omega\sqrt{n+1}$

$a\,|n\rangle = \sqrt{n}\,|n-1\rangle$; $a^\dagger\,|n\rangle = \sqrt{n+1}\,|n+1\rangle$. Therefore $|g,0\rangle$ is *dark to red-sideband excitation*, but $|e,0\rangle \leftrightarrow |g,1\rangle$ couples.

B. 2025 Q2: $\pi_A \to 2\pi_B \to \pi_A$

Let $|e\rangle = |1\rangle$, $|g\rangle = |0\rangle$, initial $n = 0$. Ion A: red sideband $g \leftrightarrow e$; ion B: auxiliary $g \leftrightarrow a$. The active π step multiplies by $-i$; the active 2π cycle multiplies by -1.

Input	after π_A	after $2\pi_B$	final π_A
$gg,0$	$gg,0$	$gg,0$	$gg,0$
$ge,0$	$ge,0$	$ge,0$	$ge,0$
$eg,0$	$-igg,1$	$+igg,1$	$+eg,0$
$ee,0$	$-ige,1$	$-ige,1$	$-ee,0$

$$\boxed{U = \text{diag}(1,1,1,-1) = \text{CZ}.}$$

Method: the B 2π pulse is active *only when* a phonon exists and $B = g$. Every branch ends with $n = 0$; put its final phase on the diagonal.

Extension: 2025 Q2B, a CCZ uses five pulses $\pi_A \to \pi_B \to 2\pi_C \to \pi_B \to \pi_A$, so only $|111\rangle$ acquires -1. A Toffoli is H on the target before/after CCZ.

C. 2023 Q1: the Mølmer–Sørensen mechanism

Two virtual paths exchange phonons with opposite detunings $\pm\delta$. Using the problem's second-order rate $\Omega_{2\gamma} = \Omega_1\Omega_2/(2\delta)$ and summing both ion orderings,

$$\Omega_{n+1} = \frac{\eta^2\Omega_r\Omega_b(n+1)}{\delta}, \quad \Omega_n = -\frac{\eta^2\Omega_r\Omega_b n}{\delta}.$$

$$\boxed{\Omega_{\text{eff}} = |\Omega_{n+1} + \Omega_n| = \frac{\eta^2|\Omega_r\Omega_b|}{|\delta|}.}$$

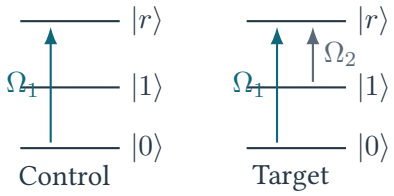
The n-dependent pieces cancel: $(n+1) - n = 1$. A Bell state from $|gg\rangle$ requires a $\pi/2$ *effective* pulse:

$$\boxed{t_{\text{Bell}} = \pi/(2\Omega_{\text{eff}}).}$$

This relies on virtual intermediate states ($|\delta| \gg$ sideband coupling), correct relative detuning signs, and laser-phase calibration.

D. 2024 Q3: simultaneous Rydberg drives

Levels: the control has $\Omega_1 : |0\rangle \leftrightarrow |r\rangle$; the target also receives $\Omega_2 : |1\rangle \leftrightarrow |r\rangle$. Simultaneous two-colour target driving can swap $|0\rangle \leftrightarrow |1\rangle$ via $|r\rangle$ *unless* the control is Rydberg excited.



Blockade: $|U_{rr}|/\hbar \gg \Omega_1, \Omega_2$ shifts the target's intermediate level off resonance when the control is $|r\rangle$.

Input	Target flip?	Output (ignoring phases)
$\|1_c 0_t\rangle$	yes	$\|1_c 1_t\rangle$
$\|1_c 1_t\rangle$	yes	$\|1_c 0_t\rangle$
$\|0_c 0_t\rangle$	blocked	$\|0_c 0_t\rangle$
$\|0_c 1_t\rangle$	blocked	$\|0_c 1_t\rangle$

Pulse method: engineer a *closed* 2π control excursion and a target *effective* π bit flip in the unblocked branch. For equal resonant target couplings, the physical 3-level *bright-state* cycle is 2π to implement the swap ($-X$), not a naive direct two-level π pulse. Exact durations need the coupling ratio/detuning/envelopes.

E. Extend the geometry to Toffoli

Place controls at opposite sides of a central target ($c_1 - t - c_2$) so each control can blockade the target, but controls do not mutually blockade. Only $c_1 = c_2 = 1$ leaves the target unblocked:

$$|c_1 c_2 t\rangle \mapsto |c_1 c_2,\ t \oplus (c_1 c_2)\rangle\,.$$

Check: identify any local phases from completed Rydberg cycles and compensate if the task requires the *exact* gate, rather than the classical truth table.

Key distinction: MS motion is approximately *virtually* excited and need not start in $n = 0$; the 2025 resonant three-pulse gate explicitly uses a real phonon and starts in the motional ground state.

4. Photons, teleportation and state preparation

2024 Q2; 2025 Q3; 2026 HOM

Photonics rule: choose the encoding first. *Dual-rail:* one photon distributed between two modes, $|0_L\rangle = |1,0\rangle$ and $|1_L\rangle = |0,1\rangle$. *Single-rail:* vacuum $|0\rangle$ vs one photon $|1\rangle$ in one mode. The 2024 paper switches encoding between its two parts.

A. Beam-splitter matrix from creation operators

2024 Q2A. If the given mapping is

$$a^\dagger \mapsto a^\dagger \cos\theta + b^\dagger e^{i\phi}\sin\theta,$$
$$b^\dagger \mapsto b^\dagger \cos\theta - a^\dagger e^{-i\phi}\sin\theta,$$

apply to $a^\dagger|\text{vac}\rangle$ and $b^\dagger|\text{vac}\rangle$. Place resulting coefficients as columns:

$$B(\theta,\phi) = \begin{pmatrix} \cos\theta & -e^{-i\phi}\sin\theta \\ e^{i\phi}\sin\theta & \cos\theta \end{pmatrix}.$$

For $\theta = \pi/4$, $\phi = 0$: $B = HZ$, so an *input* π *phase shift* on the second rail gives exactly

$$\boxed{H = B(\pi/4, 0)Z.}$$

Check matrix column order and phase convention: a bare beam splitter of this form has equal diagonal entries, so cannot equal the standard H alone.

B. Hong–Ou–Mandel: two identical photons

For a balanced splitter in a convenient phase convention,

$$a^\dagger \mapsto (c^\dagger + d^\dagger)/\sqrt{2}, \quad b^\dagger \mapsto (c^\dagger - d^\dagger)/\sqrt{2}.$$

Apply *both* operators to $|1,1\rangle = a^\dagger b^\dagger|\text{vac}\rangle$:

$$|1,1\rangle \mapsto \frac{(c^\dagger)^2 - (d^\dagger)^2}{2}|\text{vac}\rangle = \frac{|2,0\rangle - |0,2\rangle}{\sqrt{2}}.$$

Check: no $|1,1\rangle$ coincidence term. The two alternatives cancel when photons are indistinguishable; distinguishability destroys that cancellation. Use $|2,0\rangle = (c^\dagger)^2|\text{vac}\rangle/\sqrt{2}$ to get the factors right.

C. 2024 Q2B: heralded single-rail teleportation

Input in mode 0: $|\psi\rangle_0 = \alpha|0\rangle + \beta|1\rangle$; resource across modes 1–2:

$$|t\rangle_{12} = \frac{|0,1\rangle + |1,0\rangle}{\sqrt{2}}.$$

Expand $(\alpha|0\rangle + \beta|1\rangle)\otimes|t\rangle$. **Project first** onto $n_0 + n_1 = 1$ at Alice's detectors. The unnormalised surviving state is

$$|\Psi_1\rangle = \frac{\alpha|0,1\rangle_{01}|0\rangle_2 + \beta|1,0\rangle_{01}|1\rangle_2}{\sqrt{2}}.$$

$\|\Psi_1\|^2 = (|\alpha|^2 + |\beta|^2)/2 = 1/2$. After Alice's splitter, each one-click detector pattern has probability $1/4$:

(n_0, n_1)	Bob's conditional state	Correction
$(0,1)$	$\alpha\|0\rangle + \beta\|1\rangle$	I
$(1,0)$	$\alpha\|0\rangle - \beta\|1\rangle$	Z

$$\boxed{P_{\text{success}} = \tfrac{1}{4} + \tfrac{1}{4} = \tfrac{1}{2}.}$$

Zero or two counts are failure events, not alternate recoveries.

D. Standard qubit teleportation

Alice possesses an *unknown* $|\psi\rangle$ and her half of a Bell pair; Bob has the other half. Procedure:

1. Alice performs a Bell measurement on the unknown qubit and her resource qubit.
2. Send its *two classical bits* to Bob.
3. Bob applies the corresponding I, X, Z, XZ correction (depending on Bell convention).

The input is consumed (no cloning). Bob's unconditioned state cannot reveal Alice's outcome: no superluminal signalling.

E. 2025 Q3: equatorial remote preparation

State known to Alice, singlet shared with Bob. Target and axis are

$$|\beta\rangle = \frac{|0\rangle + e^{i\phi}|1\rangle}{\sqrt{2}}, \quad \boldsymbol{n} = (-\sin\phi, \cos\phi, 0).$$

Alice acts with $R_{\boldsymbol{n}}(\pi/2)$ then measures Z:

$$R = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & -e^{-i\phi} \\ e^{i\phi} & 1 \end{pmatrix}.$$

Apply $(R\otimes I)$ to $|\Psi^-\rangle = (|01\rangle - |10\rangle)/\sqrt{2}$; group by Alice's $|0\rangle, |1\rangle$ and normalise Bob's states:

$$\boxed{m = 0: \ |\beta\rangle, \qquad m = 1: \ Z|\beta\rangle}$$

(up to global phases; each probability $1/2$). Alice sends *one bit*; Bob corrects with Z^m. Polarisation implementation: entangled-photon source → Alice waveplates → PBS and two detectors → classical bit → Bob phase plate.

Arbitrary target $|\beta'\rangle = \cos(\theta/2)|0\rangle + e^{i\phi}\sin(\theta/2)|1\rangle$: Alice also controls θ via waveplates. The other measurement outcome is orthogonal and *cannot* be corrected by a single universal Z for all θ, ϕ; accept a heralded $1/2$ branch or use more general feed-forward/protocol resources.

F. CV teleportation (2025 Q3B)

Two-mode squeezing supplies quadrature entanglement; beam splitter + two homodyne measurements provide x, p results; Bob uses classical feed-forward displacement. **Strength:** efficient detectors, often near-deterministic Gaussian operation. **Limitation:** finite squeezing adds noise; optical loss and finite bandwidth reduce fidelity. For ideal unity-gain coherent-state teleportation, $F = 1/(1 + e^{-2r})$ (squeezing r).

Three checks: photon-number conservation in a passive splitter; normalise $|2\rangle$ with $\sqrt{2!}$; distinguish a *known* state (RSP) from an *unknown* state (teleportation).

5. Noise, circuit quantisation and lookup map

2026 assignments + full past-paper index

A. Quantum channels: input $\rho \to$ output ρ'

$$\boxed{\mathcal{E}(\rho) = \sum_j K_j \rho K_j^\dagger, \qquad \sum_j K_j^\dagger K_j = \mathbb{I}.}$$

Procedure: write ρ in the stated basis; compute each $K_j \rho K_j^\dagger$; add; check trace 1, positivity and limiting cases ($p = 0, p = 1$). For energy relaxation $|1\rangle \to |0\rangle$ with probability p,

$$K_0 = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-p} \end{pmatrix}, \quad K_1 = \begin{pmatrix} 0 & \sqrt{p} \\ 0 & 0 \end{pmatrix}.$$

In Bloch form, amplitude damping maps

$$\boxed{(r_x, r_y, r_z) \mapsto (\sqrt{1-p}\, r_x, \sqrt{1-p}\, r_y, p + (1-p) r_z).}$$

Pure dephasing instead reduces r_x, r_y but leaves r_z fixed (when no driving). Unitary gates *rotate* the Bloch ball; channels may contract/translate it.

B. Lindblad master equations (2026 A2)

$$\boxed{\dot\rho = -\frac{i}{\hbar}[H, \rho] + \sum_j \gamma_j \mathcal{D}[L_j]\rho},$$

$$\mathcal{D}[L]\rho = L\rho L^\dagger - \tfrac{1}{2}\{L^\dagger L, \rho\}.$$

Method: (1) expand $\rho = (I + r_x X + r_y Y + r_z Z)/2$; (2) calculate commutator and dissipators using Pauli products; (3) collect X, Y, Z coefficients; (4) solve $\dot{\boldsymbol{r}} = A\boldsymbol{r} + \boldsymbol{b}$; (5) find steady state from $A\boldsymbol{r}_{\rm ss} + \boldsymbol{b} = 0$; check purity.

For $\dot\rho = -i\Omega[X, \rho]/2 + \gamma\mathcal{D}[Z]\rho$:

$$\boxed{\begin{aligned} \dot r_x &= -2\gamma r_x, \\ \dot r_y &= -\Omega r_z - 2\gamma r_y, \\ \dot r_z &= \Omega r_y. \end{aligned}}$$

If $\Omega, \gamma > 0$, $\boldsymbol{r}_{\rm ss} = 0$, so $\rho_{\rm ss} = I/2$. If $\Omega = 0$ (only dephasing), r_z is conserved: there is *not* a unique steady state. A conventional decoherence relation is $1/T_2 = 1/(2T_1) + 1/T_\phi$.

C. Cooper-pair box: derive, do not memorise only

2026 A3: a capacitor–Josephson circuit. Use flux Φ as coordinate; voltage $\dot\Phi$. The Lagrangian is

$$\mathcal{L} = \tfrac{1}{2} C_0 \dot\Phi^2 + \tfrac{1}{2} C_c (\dot\Phi - V_g)^2 + E_J \cos\varphi, \quad \varphi = 2\pi\Phi/\Phi_0.$$

Steps: (1) compute $Q = \partial\mathcal{L}/\partial\dot\Phi = C_\Sigma \dot\Phi - C_c V_g$; (2) solve for $\dot\Phi$; (3) Legendre transform $H = Q\dot\Phi - \mathcal{L}$; (4) quantise $[\Phi, Q] = i\hbar$.

$$\boxed{H = \frac{(Q + C_c V_g)^2}{2C_\Sigma} - E_J \cos\varphi, \quad C_\Sigma = C_0 + C_c.}$$

With $E_C = e^2/(2C_\Sigma)$ and $n_g = -C_c V_g/(2e)$,

$$H = 4E_C (n - n_g)^2 - E_J \cos\varphi.$$

Expand $\cos\varphi \simeq 1 - \varphi^2/2$ near the well minimum:

$$\boxed{\hbar\omega_p \simeq \sqrt{8 E_J E_C}.}$$

The quartic correction gives the anharmonicity that isolates the qubit levels; large E_J/E_C reduces charge sensitivity but reduces anharmonicity.

D. At a glance: question-to-method index

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E. Five checks before submitting an answer

1. **Basis:** specify order ($|00\rangle, |01\rangle, |10\rangle, |11\rangle$, and motional/auxiliary labels if relevant).
2. **Conventions:** are lasers/on-0 controls and beam-splitter signs clearly declared? Is an ignored phase truly *global*?
3. **Normalization:** $\sum p = 1$, $\operatorname{Tr}\rho = 1$, $U^\dagger U = I$; for photons include Fock factorials.
4. **Reasoning:** explain why each physical transition is allowed, dark or blocked. For QEC, explain why equal syndromes are harmless or harmful.
5. **Answer the exact demand:** "detect" vs "correct", "gate up to phases" vs exact matrix, "pulse area" vs pulse time, and "known" vs unknown state.

F. Useful mathematical one-liners

$$XY = iZ, \quad YZ = iX, \quad ZX = iY, \quad \{X, Z\} = 0.$$

$$|\psi\rangle \to U|\psi\rangle, \quad \rho \to U\rho U^\dagger, \quad F = \langle\psi|\rho|\psi\rangle.$$

$$|\psi\rangle_{AB} \text{ entangled} \not\Rightarrow \rho_A \text{ pure}.$$

$$\text{QEC: } [[n, k, d]], \quad k = n - r, \quad t = \lfloor (d-1)/2 \rfloor.$$

This sheet is a navigation tool, not a substitute for full derivations. For unfamiliar signs or detailed physical assumptions, consult the companion *Part II Worked Examples* and *2023–2025 Fully Worked Past Exams* in the same repository.